目录

1 考点 1: 利用期限结构模型进行套利定价

1. A risk analyst at a financial institution is preparing a report on the key aspects of risk and risk management to present at an upcoming conference. Which of the following statements regarding risk and risk management is correct?

A. Investors typically care more about positive surprises (unexpected investment gains) than negative outcomes (unexpected investment losses).

B. Investors focus more on unexpected losses rather than expected losses when managing risk.

C. Investors pay close attention to potential losses during the risk management process, as risk is related to the magnitude of potential losses.

D. Investors should not view risk management as a zero-sum game, as the process can eliminate risk over time.

答案:B
解析:
A 选项错误。

- 虽然投资者通常关注意外的投资收益，但在风险管理中，他们更关心的是如何避免意外的投资损失。
- 心理学研究表明，人们对负面信息的关注度通常高于正面信息，这种现象被称为" 负面偏见"。

B 选项正确。

- 投资者在进行风险管理时确实更关注意外损失，而不是预期损失。这是因为：
  - 意外损失可能对投资组合造成重大影响
  - 预期损失通常已经反映在资产价格中
  - 风险管理的主要目标是防范意外风险

C 选项错误。

- 虽然投资者关注潜在损失，但风险的大小并不总是与潜在损失的大小成正比。许多大额损失是可预测的，可以通过风险管理技术进行有效管理。

D 选项错误。

- 风险管理本质上是一个风险重新分配的过程，而不是消除风险的过程。风险可能在不同参与者之间转移，但整体风险并未消除。

考察: 理解风险管理：更关注意外损失而非预期损失

2. Regarding the pricing of fixed-income investments with binomial trees, a risk-neutral approach assumes that the value of the asset:

A. Remains constant over time

B. Follows a random walk process

C. Grows at the risk-free rate

D. Grows at the rate of expected returns

答案:C
解析:
A 选项错误。

- 在风险中性定价中，资产价值会随时间变化，而不是保持不变。

B 选项错误。

- 随机游走过程不能准确反映风险中性定价的假设，因为风险中性定价要求资产以无风险利率增长。

C 选项正确。

- 风险中性定价方法假设：
  - 资产价值以无风险利率增长
  - 投资者对风险持中性态度
  - 所有资产都按无风险利率折现

D 选项错误。

- 预期收益率包含了风险溢价，而风险中性定价要求使用无风险利率。

考察： 理解风险中性定价：资产以无风险利率增长

3. A risk-free zero-coupon bond with a face value of USD 1,000 and a maturity of one year is being traded at a price of USD 945. The current risk-free 6-month interest rate is 5.5%. An analyst uses a simple two-period risk-neutral binomial pricing model that assumes the risk-free 6-month interest rate in 6 months will be either 6.0% or 5.0%. The risk-neutral probability model suggests that:

A. The 6-month risk-free interest rate is equally likely to decrease to 5.0%

B. The 6-month risk-free interest rate is more likely to decrease to 5.0% than to increase to 6.0%.

C. The 6-month risk-free interest rate is less likely to decrease to 5.0% than to increase to 6.0%.

D. There is insufficient information to determine the risk-neutral probabilities.

答案:C
解析:
步骤 1：确定参数

- 债券面值：$1,000
- 当前价格：$945
- 当前 6 个月利率：$r_0 = 5.5\%$
- 6 个月后可能的利率：$r_u = 6.0\%$（上升）或 $r_d = 5.0\%$（下降）
- 风险中性概率：p（上升概率），$1 - p$（下降概率）

步骤 2：计算 6 个月后的债券价值

- 如果利率上升到 6.0%：
$$V_u = \frac{1{,}000}{1 + \frac{6.0\%}{2}} = \frac{1{,}000}{1.03} \approx 970.87$$
- 如果利率下降到 5.0%：
$$V_d = \frac{1{,}000}{1 + \frac{5.0\%}{2}} = \frac{1{,}000}{1.025} \approx 975.61$$

步骤 3：应用风险中性定价

- 当前价格应等于 6 个月后期望价值的折现值：
$$945 = \frac{p \times V_u + (1 - p) \times V_d}{1 + \frac{r_0}{2}}$$
- 代入数值：
$$945 = \frac{p \times 970.87 + (1 - p) \times 975.61}{1 + \frac{5.5\%}{2}} = \frac{p \times 970.87 + (1 - p) \times 975.61}{1.0275}$$

- 求解风险中性概率

$$p \approx 0.85 = 85\%$$

结论

- 利率上升到 6.0% 的风险中性概率为 85%，下降到 5.0% 的概率为 15%。
- 因此，利率更可能上升而非下降。

考察： 理解风险中性定价：根据当前价格计算概率

4. A European put option has two years to expiration and a strike price of $102.00. The underlying is a 6% annual coupon bond with three years to maturity. Assume that the riskneutral probability of an up move is 0.78 in year 1 and 0.65 in year 2. The current interest rate is 3.50%. At the end of year 1, the rate will either be 6.25% or 4.75%. If the rate in year 1 is 6.25%, it will either rise to 8.75% or rise to 6.50% in year 2. If the rate in one year is 4.75%, it will either rise to 6.50% or rise to 5.00%. The value of the put option today is closest to:

A. $1.25.

B. $1.45.

C. $1.65.

D. $2.05.

答案:A
解析:
步骤 1：确定参数

- 欧式看跌期权：执行价 K = $102，2 年到期
- 标的债券：3 年期，年息 6%，面值假设为 $100
- 当前利率：$r_0 = 3.50\%$
- 风险中性概率：第 1 年上升 $p_1 = 0.78$，第 2 年上升 $p_2 = 0.65$

步骤 2：计算第二年末各节点的债券价格和期权价值

- 第二年末债券还有 1 年到期，需支付本金和最后一期利息
- 如果利率为 8.75%：债券价格 $= \frac{100+6}{1.0875} = 97.47$，期权价值 $= \max(102 - 97.47, 0) = \$4.53$
- 如果利率为 6.50%：债券价格 $= \frac{100+6}{1.065} = 99.53$，期权价值 $= \max(102 - 99.53, 0) = \$2.47$
- 如果利率为 5.00%：债券价格 $= \frac{100+6}{1.05} = 100.95$，期权价值 $= \max(102 - 100.95, 0) = \$1.05$
- 注意：解析中给出的数值可能有四舍五入差异

步骤 3：计算第一年末上层节点（利率 6.25%）的期权价值

- 第二年末可能状态：
  - 利率 8.75%：期权价值 $2.65（解析中数值）
  - 利率 6.50%：期权价值 $0.45（解析中数值）
- 第一年末期权价值：

$$V_{1u} = \frac{0.65 \times 2.65 + 0.35 \times 0.45}{1.0625} = \frac{1.7225 + 0.1575}{1.0625} = \frac{1.88}{1.0625} \approx \$1.68$$

步骤 4：计算第一年末下层节点（利率 4.75%）的期权价值

- 第二年末可能状态：
  - 利率 6.50%：期权价值 $0.42（解析中数值）

– 利率 5.00%：期权价值 \$0.00

- 第一年末期权价值：

$$V_{1d} = \frac{0.65 \times 0.42 + 0.35 \times 0.00}{1.0475} = \frac{0.273}{1.0475} \approx \$0.26$$

步骤 5：计算当前期权价值

- 当前期权价值：

$$V_0 = \frac{0.78 \times 1.68 + 0.22 \times 0.26}{1.0350} = \frac{1.3104 + 0.0572}{1.0350} = \frac{1.3676}{1.0350} \approx \$1.25$$

考察： 掌握二叉树定价：根据风险中性概率计算期权价值

5. The Black-Scholes-Merton option pricing model is not appropriate for valuing options on government bonds because government bonds:

A. have interest rate risk.

B. have an upper price bound.

C. have constant price volatility.

D. are not priced by arbitrage.

答案:B
解析:
A 选项错误。

- 利率风险不是 BSM 模型不适用于政府债券的主要原因，因为 BSM 模型可以处理标的资产的风险。

B 选项正确。

- BSM 模型不适用于政府债券定价，因为其假设不适用于固定收益证券：
  – 假设价格没有上限，但债券价格有上限（面值加上所有未付利息）
  – 假设无风险利率不变，但利率会随时间变化
  – 假设波动率不变，但债券波动率会随到期日临近而下降

C 选项错误。

- 债券价格波动率不是常数，会随到期日临近而下降。

D 选项错误。

- 政府债券可以通过无套利定价，这不是 BSM 模型不适用的主要原因。

考察： 理解 BSM 模型局限：债券价格有上限，波动率非常数

6. Which of the following statements is/are correct regarding the pricing of fixed-income securities? I. The call price serves as the ceiling value for a callable bond when interest rates fall. II. The more frequent the time steps used for projected fixed-income security values, the more accurate the pricing. III. The Black-Scholes-Merton model is preferred for the pricing of fixed income securities because of its continuous time assumption

A. II only

B. I and II only

C. I and III only

D. I, II, and III

答案:B
解析:
陈述 I：正确。

- 可赎回债券允许发行人在利率下降时以赎回价格提前赎回。
- 当利率下降、债券价格上涨时，债券价格不会超过赎回价格（因为发行人会在价格达到赎回价格时赎回）。
- 因此，赎回价格确实作为可赎回债券价格的上限。

陈述 II：正确。

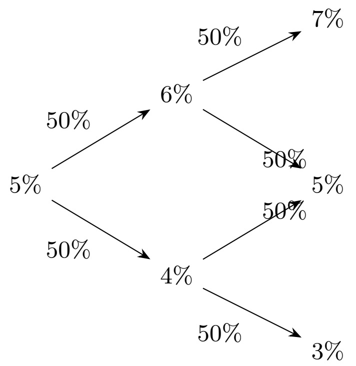
- 在二叉树等离散定价模型中，时间步长越短（步数越多），对利率路径和债券价格的模拟越精细。
- 这能更好地捕捉利率变化对债券价值的影响，提高定价精度。
- 类似于数值积分，步长越小，近似误差越小。

陈述 III：错误。

- BSM 模型不适合固定收益证券定价，主要原因：
  - BSM 假设标的资产价格没有上限，但债券价格有上限（面值加利息）
  - BSM 假设无风险利率为常数，但利率是影响债券价格的核心变量
  - BSM 假设波动率为常数，但债券价格波动率随到期时间变化
  - 固定收益证券更适合使用利率模型（如二叉树、Hull-White、Vasicek 等）

考察： 理解固定收益定价：可赎回债券有上限，时间间隔影响精度

7. Suppose investors have interest rate expectations as illustrated in the decision tree below where the 1-year rate is expected to be 7%, 5%, or 3% in the second year and either 6% or 4% in the first year for a zero-coupon bond.



If investors are risk-neutral, what is the price of a $1 face value 2-year zero-coupon bond today?

A. $0.90123.

B. $0.90567.

C. $0.90789.

D. $0.90832

答案:C
解析:
C 选项正确。

- 根据风险中性定价模型：

$$
\begin{aligned}
p &= \left(\frac{1}{1.06} \times 0.5 + \frac{1}{1.04} \times 0.5\right) \times 1.05 \\
&= 0.90789
\end{aligned}
$$

- 计算过程：
  - 第一年：利率为 6% 和 4% 的概率各为 50%
  - 第二年：利率为 7%、5%、3% 的概率各为 25%、50%、25%

– 使用当前利率 5% 进行折现

考察： 掌握二叉树定价：考虑所有路径的概率和折现

8. A binomial interest-rate tree indicates a 1-year spot rate of 5% and the price of the bond if rates decline is 96.50 and 94.25 if rates increase. The risk-neutral probability of an interest rate increase is 0.60. You hold a call option on the bond that expires in one year and has an exercise price of 94.00. The option value is closest to:

A. 1.25.

B. 1.05.

C. 1.52.

D. 1.45.

答案:D
解析:
步骤 1：确定参数

- 看涨期权：执行价 $K = 94.00$，1 年到期
- 1 年期即期利率：$r = 5\%$
- 风险中性概率：利率上升 $p = 0.60$，利率下降 $1 - p = 0.40$
- 债券价格：利率上升时 $P_u = 94.25$，利率下降时 $P_d = 96.50$

步骤 2：计算各节点的期权收益

- 利率上升时：$C_u = \max(P_u - K, 0) = \max(94.25 - 94.00, 0) = 0.25$
- 利率下降时：$C_d = \max(P_d - K, 0) = \max(96.50 - 94.00, 0) = 2.50$

步骤 3：计算风险中性下的期望收益并折现

- 期权价值：
$$C_0 = \frac{p \times C_u + (1-p) \times C_d}{1+r} = \frac{0.60 \times 0.25 + 0.40 \times 2.50}{1.05} = \frac{1.15}{1.05} \approx 1.45$$

考察： 掌握期权定价：根据风险中性概率计算期望收益

9. Suppose an investor expects that the 1-year rate will remain at 5% for the first year for a 2-year zero-coupon bond. The investor also projects a 50% probability that the 1-year spot rate will be 7% in one year and a 50% probability that the 1-year spot rate will be 3% in one year. Which of the following inequalities most accurately reflects the convexity effect for this 2-year bond using Jensen's inequality formula?

A. $\$0.95238 > \$0.95200$

B. $\$0.95200 > \$0.95000$

C. $\$0.90703 > \$0.90667$

D. $\$0.90703 > \$0.90690$

答案:C
解析:
步骤 1：理解凸性效应和 Jensen 不等式

- 函数 $f(r) = \frac{1}{1+r}$ 是凸函数（因为 $f''(r) > 0$）。
- Jensen 不等式：对于凸函数，$\mathbb{E}[f(X)] > f(\mathbb{E}[X])$。
- 因此：$\mathbb{E}\left[\frac{1}{1+r}\right] > \frac{1}{\mathbb{E}[1+r]}$。

步骤 2：计算期望折现因子

- 方法 1（正确）：
$$\mathbb{E}\left[\frac{1}{1+r}\right]=0.5\times\frac{1}{1.07}+0.5\times\frac{1}{1.03}=0.5\times 0.93458+0.5\times 0.97087=0.95273$$
- 方法 2（错误，忽略凸性）：
$$\frac{1}{\mathbb{E}[1+r]}=\frac{1}{0.5\times 1.07+0.5\times 1.03}=\frac{1}{1.05}=0.95238$$

步骤 3：计算 2 年期债券价格

- 使用正确方法：
$$P_{\text{正确}}=\frac{0.95273}{1.05}=0.90703$$
- 使用错误方法：
$$P_{\text{错误}}=\frac{0.95238}{1.05}=0.90667$$

结论

- 凸性效应导致：$0.90703>0.90667$。
- 正确考虑利率不确定性时，债券价格更高，体现了凸性的价值。

考察： 理解凸性效应：Jensen 不等式导致期望折现大于折现期望

10. Which of the following statements about puttable bonds compared to non-puttable bonds is false?

A. They have less price volatility.

B. They have positive convexity.

C. Capital losses are capped as yields fall.

D. At high yields, reinvestment rate risk rises.

答案:D
解析:
A 选项错误。

- 可回售债券确实具有较小的价格波动性，因为回售权限制了价格下跌。

B 选项错误。

- 可回售债券确实具有正凸性，因为回售权在利率上升时提供了额外的保护。

C 选项错误。

- 可回售债券确实在收益率下降时限制了资本损失，因为投资者可以按回售价格卖出债券。

D 选项正确。

- 可回售债券具有以下特征：
  - 价格波动较小
  - 正凸性
  - 收益率下降时资本损失有限
  - 收益率上升时再投资风险增加（而不是下降时）

考察： 理解可回售债券：具有正凸性，限制下跌风险

11. Suppose a stock price is currently \$60.00 while the risk-free rate is 4.0% per annum. For a series of options with a three month (0.25 years) maturity, the implied volatility is calculated at various strike prices, as below:

- Stock Price, S(0): \$60.00
- Risk-free rate: 4.0%
- Maturity (years): 0.25

A. The market price (trades) of the put is \$4.20 and this is lower than the BSM model-based price which assumes an (ATM) volatility of 65.5%

B. The market price (trades) of the put is \$4.20 and this is higher than the BSM model-based price which assumes an (ATM) volatility of 65.5%

C. The market price (trades) of the put is \$5.25 and this is lower than the BSM model-based price which assumes an (ATM) volatility of 65.5%

D. The market price (trades) of the put is \$5.25 and this is equal to the BSM model-based price which assumes an (ATM) volatility of 65.5%

答案:A
解析:
A 选项正确。

- 根据看跌期权平价公式：

$$\begin{aligned} p &= c + Ke^{-rt} - S \\ &= 8.75 + 55e^{-4\% \times 0.25} - 60 \\ &= 4.20 \end{aligned}$$

更高的波动率意味着更高的期权价格，因此市场价格低于 BSM 模型价格。

考察： 理解期权定价：波动率越高，期权价格越高

12. Which of the following factors is (are) reason(s) why the Black-Scholes-Merton option pricing model is not appropriate for valuing options on government bonds?
I. Bonds have interest rate risk.
II. Have an upper price bound.
III. Bonds do not have constant price volatility.
IV. Bonds are not priced by arbitrage.

A. I and II.

B. II and III.

C. III and IV.

D. IV only.

答案:B
解析:
陈述 I：不是主要原因。

- 虽然债券有利率风险，但 BSM 模型可以处理标的资产的风险。
- BSM 模型假设无风险利率为常数，但这不是其不适用于债券期权的主要原因。

陈述 II：正确。

- 债券价格存在上限：面值加上所有未付利息（到期时最多等于面值 + 利息总和）。
- BSM 模型假设标的资产价格可以无限增长，这与债券价格有上限的特征不符。

陈述 III：正确。

- 债券价格波动率不是常数：随着到期时间临近，债券价格波动率会下降（因为价格收敛到面值）。
- BSM 模型假设波动率为常数，这与债券波动率的时变特征不符。

陈述 IV：错误。

- 政府债券及其期权都可以通过无套利定价。
- 无套利定价是金融定价的基本原则，并非 BSM 模型不适用的原因。

考察： 理解 BSM 模型局限：债券价格有上限，波动率非常数

13. A European put option has two years to expiration and a strike price of \$101.00. The underlying asset is a three-year bond with a 7% coupon rate. Assume that the risk-neutral probability of an up move is 0.76 in year 1 and 0.60 in year 2. The current interest rate is 3.00%. At the end of year 1, the rate will either be 5.99% or 4.44%. If the rate in year 1 is 5.99%, it will either rise to 8.56% or rise to 6.34% in year 2. If the rate in year 1 is 4.44%, it will either rise to 6.34% or rise to 4.70%. The current value of the put option is closest to:

A. \$1.17

B. \$1.30

C. \$1.49

D. \$1.98

答案:A
解析:
步骤 1：计算第二年末各节点的期权价值

- 第二年末债券还有 1 年到期，需支付本金和利息 \$107
- 利率 8.56%：债券价格 \$98.56，期权价值 =\$2.44
- 利率 6.34%：债券价格 \$100.62，期权价值 =\$0.38
- 利率 4.70%：债券价格 \$102.20，期权价值 =\$0.00

步骤 2：计算第一年末各节点的期权价值

- 上层节点（利率 5.99%）：
$$V_{1u} = \frac{0.60 \times 2.44 + 0.40 \times 0.38}{1.0599} \approx \$1.52$$
- 下层节点（利率 4.44%）：
$$V_{1d} = \frac{0.60 \times 0.38 + 0.40 \times 0.00}{1.0444} \approx \$0.22$$

步骤 3：计算当前期权价值

- 当前期权价值：
$$V_0 = \frac{0.76 \times 1.52 + 0.24 \times 0.22}{1.0300} \approx \$1.17$$

考察： 掌握二叉树模型：使用风险中性概率计算期权价值

14. An analyst at an investment bank is studying interest rate trees and how they are constructed and used to price fixed-income contingent claims under the no-arbitrage assumption. The analyst focuses on the characteristics and features of interest rate trees constructed using risk-neutral pricing. Which of the following is correct?

A. Risk-neutral pricing assumes that the probabilities of both up and down states equal 0.5, reflecting a neutral view on the direction of interest rate movements.

B. Risk-neutral pricing requires creating a new interest rate tree for each fixed-income contingent claim.

C. Risk-neutral pricing makes the expected discounted value of fixed-income contingent claims equal to their market prices.

D. Risk-neutral pricing requires constructing and pricing a replicating portfolio for contingent claims, even when the risk-neutral tree for interest rates is known.

答案:C
解析:
A 选项错误。

- 风险中性概率不一定等于 0.5，它们是通过使标的证券价格等于其预期折现价值来确定的。

B 选项错误。

- 风险中性定价是一种非常高效的方法，可以在相同的假设利率过程下对多个或有权益进行定价。

C 选项正确。

- 风险中性定价的核心是找到使标的证券价格等于其预期折现价值的风险中性概率。

D 选项错误。

- 一旦已知利率的风险中性树，就可以直接对任何或有权益进行定价，无需构建和定价其复制组合。

考察: 理解风险中性定价：使预期折现价值等于市场价格

2 考点 2: 预期、风险溢价、凸性和期限结构的形状

1. If all other factors remain unchanged, what happens to the price-yield curve of a zero-coupon bond as its maturity increases?

A. The curve becomes flatter.

B. The curve becomes steeper.

C. The curve becomes less convex.

D. The curve becomes more convex.

答案:D
解析:
A 选项错误。

- 随着期限增加，价格-收益率曲线不会变得更平坦，而是凸性增强。

B 选项错误。

- 曲线变陡并不是主要特征，主要是凸性变化。

C 选项错误。

- 期限越长，凸性越大，不会变得更不凸。

D 选项正确。

- 随着零息债券期限的增加，价格-收益率曲线的凸性增强。

考察: 凸性与期限

2. Suppose an investor anticipates that the 1-year zero-coupon bond rate will stay at 6% this year, rise to 8% next year, and reach 10% in the third year. What is the 3-year spot rate for a zero-coupon bond with a face value of $1, assuming all investors share the same expectations for future 1-year rates?

A. 7.888%

B. 7.988%

C. 8.000%

D. 8.088%

答案:B
解析:
步骤 1：理解即期利率的计算

- 3 年期即期利率是将 1 年期远期利率序列按复利几何平均得到。
- 根据期望理论，3 年期零息债券的现值应等于按 1 年期远期利率序列折现的现值。

步骤 2：建立等价关系

- 按 3 年期即期利率折现：
$$\frac{\$1}{(1+r_3)^3}$$
- 按 1 年期远期利率序列折现：
$$\frac{\$1}{1.06 \times 1.08 \times 1.10}$$
- 两者相等：
$$\frac{\$1}{(1+r_3)^3} = \frac{\$1}{1.06 \times 1.08 \times 1.10}$$

步骤 3：计算 3 年期即期利率

- 整理方程：
$$(1+r_3)^3 = 1.06 \times 1.08 \times 1.10 = 1.25928$$
- 求解：
$$r_3 = \sqrt[3]{1.25928} - 1 = 1.07988 - 1 = 0.07988 = 7.988\%$$

考察: 即期利率计算

3. If all other factors remain constant, what happens to the bond price-yield curve as the maturity of a zero-coupon bond increases? The pricing curve becomes:

A. less concave.

B. more concave.

C. less convex.

D. more convex.

答案:D
解析:
A 选项错误。

- 随着期限增加，曲线不会变得更凹。

B 选项错误。

- 曲线的主要变化是凸性增强，而非凹性。

C 选项错误。

- 期限越长，凸性越大，不会变得更不凸。

D 选项正确。

- 随着零息债券期限增加，价格-收益率曲线凸性增强。

考察: 凸性与期限关系

4. A market risk manager seeks to calculate the price of a two-year zero-coupon bond. The one-year interest rate today is 10.0%. There is a 50% probability that the interest rate will be 12.0% and a 50% probability that it will be 8.0% in one year. Assuming that the risk premium of duration risk is 50 basis points each year, and that the face value is EUR 1000, which of the following should be the price of the zero-coupon bond?

A. EUR 822.98

B. EUR 826.44

C. EUR 826.72

D. EUR 921.66

答案:B
解析:
步骤 1:确定参数

- 2 年期零息债券，面值 EUR 1,000
- 当前 1 年期利率：$r_0 = 10.0\%$
- 1 年后利率：50% 概率为 12.0%，50% 概率为 8.0%
- 期限风险溢价：每年 50 个基点（0.5%）

步骤 2:计算调整后的折现率

- 第 1 年折现率：$10.0\% + 0.5\% = 10.5\%$（考虑风险溢价）
- 如果第 2 年利率为 12.0%：折现率 $= 12.0\% + 0.5\% = 12.5\%$
- 如果第 2 年利率为 8.0%：折现率 $= 8.0\% + 0.5\% = 8.5\%$

步骤 3:计算债券价格

- 使用风险中性定价，计算期望折现值：

$$\begin{aligned}
P &= \frac{1}{1.105} \times \left[0.5 \times \frac{1}{1.125} + 0.5 \times \frac{1}{1.085}\right] \times 1{,}000 \\
&= \frac{1}{1.105} \times [0.5 \times 0.8889 + 0.5 \times 0.9217] \times 1{,}000 \\
&= \frac{1}{1.105} \times 0.9053 \times 1{,}000 \\
&= 0.8264 \times 1{,}000 \\
&= \text{EUR } 826.44
\end{aligned}$$

考察: 零息债定价与风险溢价

5. A fixed-income analyst is asked to determine the price of a 3-year zero-coupon bond, given the following market conditions: current interest rate is 2%, annual interest rate volatility is 40 bps, and investors require an annual risk premium of 10 bps. What is the current price of the 3-year zero-coupon bond under these conditions?

A. 0.9396

B. 0.9410

C. 0.9415

D. 0.9424

答案:A
解析:
步骤 1：确定参数

- 3 年期零息债券，面值假设为 \$1
- 当前利率：$r = 2\% = 0.02$
- 年利率波动率：$\sigma = 40 \text{ bps} = 0.40\% = 0.004$
- 年风险溢价：$\lambda = 10 \text{ bps} = 0.10\% = 0.001$

步骤 2：计算调整后的折现率

- 考虑风险溢价后，折现率需要调整：$r_{\text{adj}} = r + \lambda = 0.02 + 0.001 = 0.021$
- 基础折现值：$P_{\text{base}} = \frac{1}{(1.021)^3} \approx 0.9415$

步骤 3：考虑凸性效应和波动率影响

- 综合调整后的折现率需要考虑波动率的平方项（方差项）：

$$r_{\text{total}} = r + \lambda + \frac{\sigma^2 \times T}{2}$$

- 对于 3 年期债券：$r_{\text{total}} = 0.02 + 0.001 + \frac{0.004^2 \times 3}{2} = 0.021 + 0.000024 = 0.021024$
- 债券价格：

$$P = \frac{1}{(1 + r_{\text{total}})^3} = \frac{1}{(1.021024)^3} \approx 0.9396$$

考察: 零息债定价与风险溢价

疑惑：这个波动率怎么用上的，这个的公式是什么。感觉不用上波动率也可以得出 A。

3 考点 3: 期限结构模型：漂移和波动性分布

1. Tuckman's Model 1 assumes zero drift and is also called the normal model. Model 2 adds a drift term. According to Tuckman, each of the following statements about these two models is correct EXCEPT:

A. A weakness of Model 1 (normal model) is that the short-term rate can become negative.

B. Model 1 implies a term structure that is perfectly flat at the current rate for all maturities, including the long-term rates.

C. Model 2 is more capable of producing an upward-sloping term structure, which is often observed.

D. Model 2 is an equilibrium model, rather than an arbitrage-free model, because no attempt is made to match the term structure closely.

答案:B
解析:
A 选项错误。

- Model 1 允许短期利率变为负值，确实是其缺陷之一。

B 选项正确。

- Model 1 只在短期内表现为平坦结构，整体上由于凸性效应会导致期限结构向下倾斜。

C 选项错误。

- Model 2 引入漂移项，更能生成现实中常见的上升期限结构。

D 选项错误。

- Model 2 为均衡模型，确实未严格拟合实际期限结构。

考察: 模型 1 与模型 2 比较

2. The Bureau of Labor Statistics has just reported an unexpected short-term increase in high-priced luxury automobiles. What is the most likely anticipated impact on a mean-reverting model of interest rates?

A. The economic information is long-lived with a low mean reversion parameter.

B. The economic information is short-lived with a low mean reversion parameter.

C. The economic information is long-lived with a high mean reversion parameter.

D. The economic information is short-lived with a high mean reversion parameter.

答案:D
解析:
A 选项错误。

- 该经济信息属于短期影响，不是长期。

B 选项错误。

- 虽然是短期信息，但均值回复参数应高而非低。

C 选项错误。

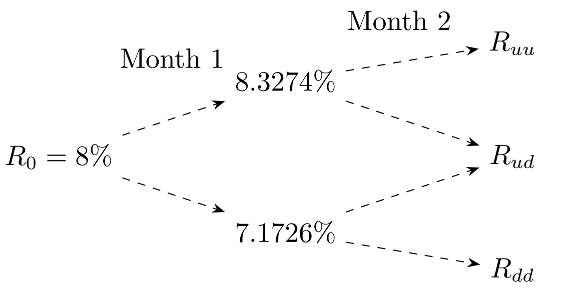
- 信息不是长期影响，均值回复参数高不适用。

D 选项正确。

- 该经济消息时效短，均值回复参数高，调整速度快。

考察: 均值回复与经济消息

3. An analyst is building a short-term interest rate tree according to the Vasicek Model which is characterized by mean reversion. t = 1/12. The current short-term rate is 8.00%. The annual volatility is 200bps. The central tendency is 3.00%. The speed of mean reversion is equal to 0.60. Here is his rate tree:



Vasicek Model (Mean-Reverting Drift)

What is the value in the Vasicek tree for Rud?

A. 6.833%

B. 7.513%

C. 8.019%

D. 9.225%

答案:B
解析:
B 选项正确。

- In month 2:

$$r_{ud} = r_u + k(\theta - r_u)dt - \sigma\sqrt{dt} = 8.3274\% + 0.6 \times \frac{3\% - 8.3274\%}{12} - 0.02 \times \sqrt{\frac{1}{12}} = 7.484\%$$

$$r_{du} = r_d + k(\theta - r_d)dt + \sigma\sqrt{dt} = 7.1726\% + 0.6 \times \frac{3\% - 7.1726\%}{12} + 0.02 \times \sqrt{\frac{1}{12}} = 7.541\%$$

$$\Rightarrow \frac{7.484\% + 7.541\%}{2} = 7.513\%$$

考察: Vasicek 模型利率树计算

4. Using Model 2, suppose the current short-term rate is 8%, the annual drift is 50bps, and the short-term rate standard deviation is 2%. If the ex-post realization of the dw random variable is 0.3, after constructing a 2-period interest rate tree with annual periods, what is the interest rate at the middle node at the end of year 2?

A. 8.0%

B. 8.8%

C. 9.0%

D. 9.6%

答案:C

解析:

步骤 1：理解 Model 2 的利率树结构

- Model 2 公式：$dr = \lambda dt + \sigma dw$
- 其中：$\lambda = 0.5\%$（年漂移），$\sigma = 2\%$（标准差），$dt = 1$ 年

步骤 2：理解中间节点的特性

- 在 2 期利率树中，第 2 年末的中间节点是两条路径的汇合点。
- 一条路径：先上升后下降（$+dw - dw$）
- 另一条路径：先下降后上升（$-dw + dw$）
- 在中间节点，随机项 dw 互相抵消，只剩下漂移项的累积效应。

步骤 3：计算中间节点的利率

- 中间节点利率公式：

$$r_{\text{mid}} = r_0 + 2 \times \lambda \times dt$$

- 代入数值：

$$r_{\text{mid}} = 8\% + 2 \times 0.5\% \times 1 = 8\% + 1\% = 9.0\%$$

- 由于 dw 项在中间节点抵消，$dw = 0.3$ 的实现在中间节点不产生影响。

考察: Model 2 利率树计算

5. An analyst is constructing an interest rate tree with monthly time steps (t = 1/12). The current short-term rate is 4.0%. The term structure model assumes an annual basis point volatility of 180bps. Using Model 1 (normally distributed rates, zero drift), what is the interest rate at the uppermost node at the end of month 2?

A. 4.20%

B. 4.83%

C. 5.04%

D. 6.59%

答案:C
解析:
C 选项正确：

- 按 Model 1，第二个月最上节点利率为：

$$r = r_0 + 2\sigma\sqrt{dt} = 4\% + 2 \times 0.018 \times \sqrt{\frac{1}{12}} \approx 5.04\%$$

- 其中，$\sigma = 180\text{bps} = 0.018$，$dt = 1/12$。两期上升路径需叠加两次标准差项，得出 5.04%。

考察：Model 1 利率树计算

6. An analyst constructed an interest rate tree with monthly time steps (t = 1/12). The current short-term rate is 3.0%. The term structure model assumes an annual basis point volatility of 200bps and an annual drift of 50bps. Using Model 2 (normally distributed rates with drift), what is the missing value at the lowest node at the end of month 2?

A. 1.93%

B. 2.17%

C. 2.38%

D. 3.01%

答案:A
解析:
A 选项正确。

- 计算最低节点利率时，需两次向下移动，每次都要减去波动项并加上漂移项。
- 计算公式为：

$$r = r_0 + 2 \times \lambda \times dt - 2 \times \sigma\sqrt{dt}$$

- 其中，$r_0 = 3.0\%$，$\lambda = 0.5\% = 0.005$，$\sigma = 2.0\% = 0.02$，$dt = 1/12$。

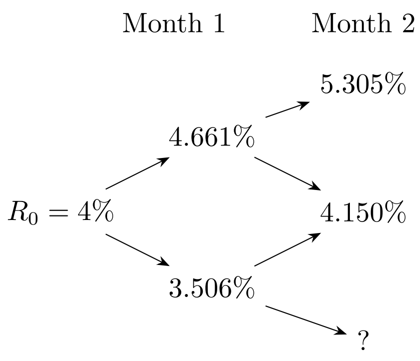
$$\begin{aligned} r &= 3.0\% + 2 \times 0.005 \times \frac{1}{12} - 2 \times 0.02 \times \sqrt{\frac{1}{12}} \\ &= 3.0\% + 0.000833 - 0.011547 \approx 1.93\% \end{aligned}$$

考察：Model 2 利率树节点计算

7. The current short-term rate is 4.00%. Under a Ho-Lee Model with time-dependent drift, the time step is monthly and the annualized drifts are as follows: +100bps in the first month and +80bps in the second month. The annual basis point volatility is 200bps.

Ho-Lee Model: Time-Dependent Drift

$$dr = \lambda_t dt + \sigma d\omega$$



What is the value of the missing node?

A. 2.447%

B. 2.677%

C. 2.995%

D. 3.256%

答案:C

解析:

步骤 1：确定参数

- 当前利率：$r_0 = 4.0\%$
- 第 1 个月漂移：$\lambda_1 = 100 \text{ bps} = 0.01$
- 第 2 个月漂移：$\lambda_2 = 80 \text{ bps} = 0.008$
- 年波动率：$\sigma = 200 \text{ bps} = 0.02$
- 时间步长：$dt = \frac{1}{12}$

步骤 2：计算 dd 节点（两期都下降）的利率

- Ho-Lee 模型：$dr = \lambda_t dt + \sigma d\omega$
- dd 节点意味着两期都下降（$-dw, -dw$），所以：

$$r_{dd} = r_0 + (\lambda_1 + \lambda_2)dt - 2\sigma\sqrt{dt}$$

- 代入数值：

$$r_{dd} = 4\% + \frac{0.01 + 0.008}{12} - 2 \times 0.02 \times \sqrt{\frac{1}{12}} = 4\% + 0.0015 - 0.01155 = 2.995\%$$

考察：Ho-Lee 模型利率树计算

8. A risk manager is pricing a 10-year call option on 10-year Treasuries using a successfully tested pricing model. Current interest rate volatility is high and the risk manager is concerned about the effect this may have on short-term rates when pricing the option. Which of the following actions would best address the potential for negative short-term interest rates to arise in the model?

A. The risk manager uses a normal distribution of interest rates.

B. When short-term rates are negative, the risk manager adjusts the risk-neutral probabilities.

C. When short-term rates are negative, the risk manager increases the volatility.

D. When short-term rates are negative, the risk manager sets the rate to zero.

答案:D

解析:

A 选项错误。

- 采用正态分布本身无法避免负利率的出现，反而可能导致负利率节点增多。

B 选项错误。

- 调整风险中性概率会影响整个利率树的动态，不仅仅局限于负利率节点，反而可能带来更多模型偏差。

C 选项错误。

- 增加波动率会使利率分布更分散，反而可能增加负利率出现的概率，不能解决根本问题。

D 选项正确。

- 当模型中出现负短期利率时，将其设为零是常用且有效的处理方法。这样可以局部修正负利率节点，同时保留其他节点的原始结构，经济含义也更合理。负利率虽然在某些模型下可能出现，但实际市场中投资者一般不会以负利率出借资金，因此将负利率设为零更符合实际情况。

考察: 负利率处理方法

9. Regarding the validity of time-dependent drift models, which of the following statements is(are) correct?
I. Time-dependent drift models are flexible since volatility from period to period can change. However, volatility must be an increasing function of short-term rate volatilities.
II. Time-dependent volatility functions are useful for pricing interest rate caps and floors.

A. I only.

B. II only.

C. Both I and II.

D. Neither I nor II.

答案:B
解析:
陈述 I：错误。

- 时间相关漂移模型允许波动率在不同时期变化，具有灵活性。但波动率不必须是短期利率波动率的递增函数。
- 波动率可以是递增、递减或恒定的，取决于模型设定和市场条件。因此，陈述 I 的后半部分（”必须递增”）是错误的。

陈述 II：正确。

- 时间相关波动率函数对利率上限和下限的定价很有用。
- 原因：
  - 利率上限和下限涉及多期支付（每期都可能触发）
  - 不同时期的波动率可能不同（如短期波动率较高，长期波动率较低）
  - 时间相关的波动率设定能更准确地捕捉不同时期的波动特征

考察: 时间相关波动率模型

10. Which of the following statements best characterizes the differences between the Ho-Lee model with drift and the lognormal model with drift?

A. In the Ho-Lee model and the lognormal model the drift terms are multiplicative.

B. In the Ho-Lee model and the lognormal model the drift terms are additive.

C. In the Ho-Lee model the drift terms are multiplicative, but in the lognormal model the drift terms are additive.

D. In the Ho-Lee model the drift terms are additive, but in the lognormal model the drift terms are multiplicative.

答案:D
解析:
A 选项错误。

- 两个模型的漂移项累积方式并非都是相乘。

B 选项错误。

- 两个模型的漂移项累积方式并非都是相加。

C 选项错误。

- Ho-Lee 模型是相加，lognormal 模型是相乘，顺序颠倒。

D 选项正确。

- Ho-Lee 模型的漂移项累积是相加的，lognormal 模型的漂移项累积是相乘的。Ho-Lee 模型允许每期漂移项变化，累积效果为加法；而对数正态模型累积效果为乘法。

考察： Ho-Lee 模型与对数正态模型

11. Lisa Wang, CFA, is discussing the appropriate usage of mean-reverting models relative to no-drift models, models that incorporate drift, and Black-Kim models. Wang makes the following statements:

- Statement 1: Both Model 1 (no drift) and the CIR model assume parallel shifts from changes in the short-term rate.
- Statement 2: The CIR model assumes decreasing volatility of future short-term rates while Model 1 assumes constant volatility of future short-term rates.
- Statement 3: The constant drift model (Model 2) is a more flexible model than the Black-Kim model.

How many of her statements are correct?

A. 0

B. 1

C. 2

D. 3

答案:B
解析:
A 选项错误：实际上有一条陈述是正确的。
B 选项正确：只有 Statement 2 是正确的。CIR 模型假设未来短期利率的波动性递减，而 Model 1 假设波动性恒定。Statement 1 错误，因为 CIR 模型不一定是平行移动；Statement 3 错误，因为 Black-Kim 模型比 Model 2 更通用，Model 2 只是 Black-Kim 模型的特例。
C 选项错误：只有一条陈述正确，不是两条。
D 选项错误：只有一条陈述正确，不是三条。
考察：利率模型特性比较

12. Using Model 1, suppose the current short-term interest rate is 5%, annual volatility is 80bps, and dw, a normally distributed random variable with mean 0 and standard deviation $\sqrt{dt}$, has an expected value of zero. After one month, the realization of dw is -0.5. What is the change in the spot rate and the new spot rate?

A. 0.4%, 5.4%

B. -0.4%, 4.6%

C. 0.8%, 5.8%

D. -0.8%, 4.2%

答案:B
解析:
A 选项错误：该选项将变化方向搞反，dw 为负时应为利率下降。
B 选项正确： Model 1 为无漂移模型，利率变化公式为 $dr = \sigma dw$。已知年化波动率为 0.8%，一个月的 dw 为-0.5，则

$$dr = 0.8\% \times (-0.5) = -0.4\%$$

初始利率为 5%，新即期利率为

$$5\% + (-0.4\%) = 4.6\%$$

计算过程和结果均正确。

C 选项错误：该选项将 dw 的符号和倍数搞错，结果偏高。

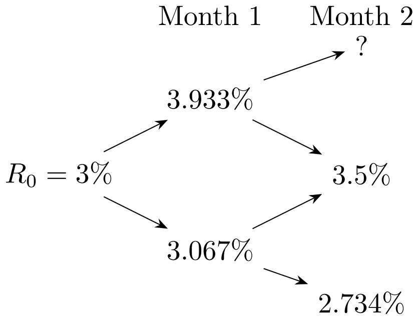
D 选项错误：该选项将 dw 的绝对值扩大，未按题意计算。

考察：无漂移模型利率变动

13. An analyst is constructing an interest rate tree with monthly time steps; i.e., $t = 1/12$. The current short-term rate is 3.5%. His term structure model assumes an annual basis point volatility of 150bps. He employs Model 1 which assumes normally distributed rates and zero drift. Here is his rate tree:

Model 1: Normally Distributed Rates, Zero Drift

$$dr = \sigma d\omega$$



What is the un-displayed missing value?

A. 3.933%

B. 4.366%

C. 4.150%

D. 4.600%

答案:B

解析:

步骤 1：确定参数

- 当前短期利率：$r_0 = 3.5\%$（题目中给出）
- 年波动率：$\sigma = 150\text{ bps} = 1.5\% = 0.015$
- 时间步长：$dt = \frac{1}{12}$（月）
- Model 1：零漂移（$\lambda = 0$），只有随机项

步骤 2：理解 uu 节点（最上节点）的特性

- uu 节点意味着两期都上升（$+dw, +dw$）。
- 每期上升幅度：$\sigma\sqrt{dt} = 0.015 \times \sqrt{\frac{1}{12}} \approx 0.015 \times 0.2887 \approx 0.00433 = 0.433\%$

步骤 3：计算 uu 节点的利率

- 最上节点利率：

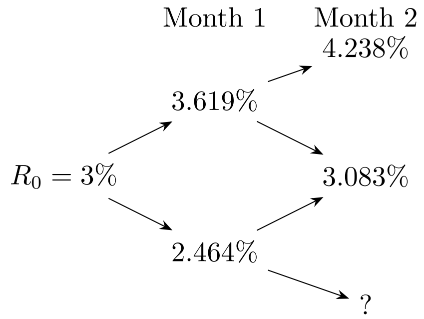
$$r_{uu} = r_0 + 2 \times \sigma\sqrt{dt} = 3.5\% + 2 \times 0.00433 = 3.5\% + 0.866\% = 4.366\%$$

考察：Model 1 利率树节点计算

14. An analyst constructed an interest rate tree with monthly time steps, where t = 1/12. The current short-term rate is 3.0%. His term structure model assumes an annual basis point volatility of 200bps with an annual drift of 50bps. He employs Model 2 which assumes normally distributed rates and incorporating drift. Here is his rate tree:

Model 2: Normally Distributed Rates but Incorporating Annual Drift

$$dr = \lambda dt + \sigma d\omega$$



What is the un-displayed missing value?

A. 1.93%

B. 2.17%

C. 2.38%

D. 3.01%

答案:A
解析:
步骤 1：确定参数

- 当前短期利率：$r_0 = 3.0\%$
- 年漂移：$\lambda = 50 \text{ bps} = 0.5\% = 0.005$
- 年波动率：$\sigma = 200 \text{ bps} = 2.0\% = 0.02$
- 时间步长：$dt = \frac{1}{12}$（月）
- Model 2：$dr = \lambda dt + \sigma d\omega$

步骤 2：理解 dd 节点（最下节点）的特性

- dd 节点意味着两期都下降（$-dw, -dw$）。
- 需要加上两期的漂移项，减去两期的随机项。

步骤 3：计算 dd 节点的利率

- 最下节点利率公式：

$$r_{dd} = r_0 + 2\lambda dt - 2\sigma\sqrt{dt}$$

- 代入数值：

$$r_{dd} = 3\% + 2 \times 0.005 \times \frac{1}{12} - 2 \times 0.02 \times \sqrt{\frac{1}{12}}$$
$$= 3\% + 0.000833 - 0.01155 = 3.000833\% - 1.155\% = 1.93\%$$

考察: Model 2 利率树节点计算

15. Using Model 2, assume a current short-term rate of 8%, an annual drift of 50bps, and a short-term rate standard deviation of 2%. In addition, assume the ex-post realization of the dw random variable is 0.3. After constructing a 2-period interest rate tree with annual periods, what is the interest rate in the middle node at the end of year 2?

A. 8.0%

B. 8.8%

C. 9.0%

D. 9.6%

答案:C
解析:
步骤 1：确定参数

- 当前短期利率：$r_0 = 8\%$
- 年漂移：$\lambda = 50 \text{ bps} = 0.5\% = 0.005$
- 标准差：$\sigma = 2\% = 0.02$
- 时间步长：$dt = 1$（年）
- Model 2：$dr = \lambda dt + \sigma d\omega$

步骤 2：理解中间节点的特性

- 在 2 期利率树中，第 2 年末的中间节点是两条路径的汇合点。
- 路径 1：先上升后下降（$+dw, -dw$）
- 路径 2：先下降后上升（$-dw, +dw$）
- 在中间节点，随机项 dw 互相抵消，只剩下漂移项的累积效应。

步骤 3：计算中间节点的利率

- 中间节点利率公式：

$$r_{\text{mid}} = r_0 + 2\lambda dt$$

- 代入数值：

$$r_{\text{mid}} = 8\% + 2 \times 0.5\% \times 1 = 8\% + 1\% = 9.0\%$$

考察: Model 2 利率树计算

16. Using Model 1, assume the current short-term interest rate is 5%, annual volatility is 80bps, and dw, a normally distributed random variable with mean 0 and standard deviation, √dt, has an expected value of zero. After one month, the realization of dw is -0.5. What is the change in the spot rate and the new spot rate?

A. 0.4%，5.4%

B. -0.4%，4.6%

C. 0.8%，5.8%

D. -0.8%，4.2%

答案:B
解析:
步骤 1：确定参数

- 当前短期利率：$r_0 = 5\%$
- 年波动率：$\sigma = 80 \text{ bps} = 0.8\% = 0.008$
- 随机变量实现：$dw = -0.5$（一个月后）
- Model 1：零漂移（$\lambda = 0$），$dr = \sigma d\omega$

步骤 2：计算利率变化

- 根据 Model 1 公式：

$$dr = \sigma \times dw = 0.8\% \times (-0.5) = -0.4\%$$

- 利率变化为-0.4%（下降 40 个基点）。

步骤 3：计算新的即期利率

- 新即期利率：

$$r_{\text{new}} = r_0 + dr = 5\% + (-0.4\%) = 4.6\%$$

考察： Model 1 利率树计算

17. Using Model 1, assume the current short-term interest rate is 5%, annual volatility is 80bps, and d , a normally distribution random variable with mean 0 and standard deviation $\sqrt{dt}$, has an expected value of zero. After one month, the realization of d is -0.5. What is the change in the spot rate and the new spot rate?

A. 0.40% （Change in Spot）；5.40%（New Spot Rate）

B. -0.40%（Change in Spot）；4.60%（New Spot Rate）

C. 0.80% （Change in Spot）；5.80%（New Spot Rate）

D. -0.80%（Change in Spot）；4.20%（New Spot Rate）

答案:B
解析:
步骤 1：确定参数

- 当前短期利率：$r_0 = 5\%$
- 年波动率：$\sigma = 80 \text{ bps} = 0.8\%$
- 随机变量实现：$d\omega = -0.5$（一个月后）
- Model 1：零漂移（$\lambda = 0$），$dr = \sigma d\omega$

步骤 2：计算利率变化

- 根据 Model 1 公式：

$$dr = \sigma \times d\omega = 0.8\% \times (-0.5) = -0.4\%$$

- 利率变化为-0.40%，即下降 40 个基点。

步骤 3：计算新的即期利率

- 新即期利率：

$$r_1 = r_0 + dr = 5\% + (-0.4\%) = 4.6\%$$

考察： Model 1 利率树计算

18. An analyst is modeling spot rate changes using short rate term structure models. The current short-term interest rate is 5% with a volatility of 80 bps. After one month passes the realization of d , a normally distributed random variable with mean 0 and standard deviation $\sqrt{dt}$, is -0.5. Assume a constant interest rate drift, , of 0.36%. What should the analyst compute as the new spot rate?

A. 5.37%

B. 4.63%

C. 5.76%

D. 4.24%

答案:B
解析:
步骤 1：确定参数

- 当前短期利率：$r_0 = 5\%$
- 年波动率：$\sigma = 80$ bps $= 0.8\%$
- 年漂移：$\lambda = 0.36\%$
- 随机变量实现：$d\omega = -0.5$（一个月后）
- 时间步长：$dt = \frac{1}{12}$（月）
- Model 2：$dr = \lambda dt + \sigma d\omega$

步骤 2：计算利率变化

- 根据 Model 2 公式：
$$dr = \lambda dt + \sigma d\omega = \frac{0.36\%}{12} + 0.8\% \times (-0.5) = 0.03\% - 0.4\% = -0.37\%$$
- 利率变化为-0.37%，即下降 37 个基点。

步骤 3：计算新的即期利率

- 新即期利率：
$$r_1 = r_0 + dr = 5\% + (-0.37\%) = 4.63\%$$

考察： Model 2 利率树计算

19. Analyst Linda Chen runs a short-term interest rate simulation using Model 1, which assumes no drift. The time step in her model is one month. Her Model 1 also makes two assumptions. First, the initial or current short-term rate is equal to 3.20%. Second, the annual basis-point volatility is 150 basis points. In the first step of her first trial, the random uniform variable is 0.9750 such that, via inverse transformation, the associated random standard normal value is 1.960. To what level does the rate evolve in the first month?

A. 2.950%

B. 3.480%

C. 3.760%

D. 4.010%

答案:B
解析:
步骤 1：确定参数

- 当前短期利率：$r_0 = 3.20\%$
- 年波动率：$\sigma = 150$ bps $= 1.5\% = 0.015$
- 时间步长：$dt = \frac{1}{12}$（月）
- 随机变量实现：$d\omega = 1.960$（标准正态分布）
- Model 1：零漂移（$\lambda = 0$），$dr = \sigma\sqrt{dt} \times d\omega$

步骤 2：计算利率变化

- 根据 Model 1 公式：
$$dr = \sigma\sqrt{dt} \times d\omega = 0.015 \times \sqrt{\frac{1}{12}} \times 1.960$$
- 计算：$dr = 0.015 \times 0.2887 \times 1.960 \approx 0.280\%$

步骤 3：计算第一个月后的利率

- 新利率：

$$r_1 = r_0 + dr = 3.20\% + 0.280\% = 3.480\%$$

考察： 无漂移模型利率模拟

20. Within the context of risk management, which of the following choices is considered most essential when assessing risks in modern financial institutions?

A. Calculating separate capital requirements for the trading and banking portfolios.

B. Comprehensive analysis of stressed Value at Risk (VaR).

C. Consistent application of backtesting procedures.

D. Taking into account the interactions among different risk factors.

答案:D

解析:

A 选项错误。

- 分开计算交易账簿和银行账簿的资本要求是监管要求，但这种方法忽略了不同类型风险之间的联动与传染效应。
- 在危机期间，这些风险往往相互放大，单独计算无法捕捉系统性风险。

B 选项错误。

- 压力 VaR 分析重要，但主要关注单一风险度量在极端场景下的表现。
- 若未建模风险因素间的交互（如市场—信用、市场—流动性），可能低估组合风险。

C 选项错误。

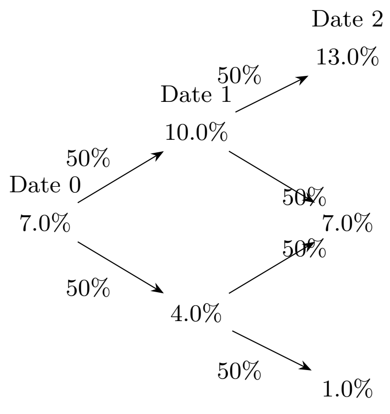
- 持续回测用于验证模型的准确性和稳定性，属于事后验证工具。
- 不能替代对风险因素间相关性、尾部依赖和结构性传染渠道的识别与建模。

D 选项正确。

- 现代金融机构面临多种风险（市场、信用、流动性、操作风险等），这些风险因素之间存在复杂的相互作用。
- 典型交互路径：
  - 市场风险 ↑ → 抵押品价值 ↓ → 流动性风险 ↑ → 被动减仓 → 市场风险进一步 ↑（流动性螺旋）
  - 利率/汇率冲击 → 债务负担恶化 → 信用利差走阔（市场—信用联动）
  - 集中度与错向风险：敞口与违约概率同向（wrong-way risk）放大尾部损失
- 只有综合考虑风险因素间的交互效应，才能更准确地评估整体风险，实现有效的风险分散和资本配置。

考察： 风险因素交互分析

21. The interest rate tree below shows the true process for a one-year interest rate. The current one-year spot rate is 7.0%. Next year, investors expect the one-year rate to either increase to 10.0% or drop to 4.0%, with equal probability of an increase or drop. In the subsequent year (Year 2), investors similarly expect the future one-year rate to again either increase or decrease by +/- 3%, with equal likelihood. Graphically, as follows:



Before the inclusion of a risk premium, assuming the above interest rate tree the true and known process, the price of a \$1,000 par two-year zero-coupon bond would be \$874.13; i.e., this price is the expected discounted value under annual compounding. However, let us modify this and instead assume that investors are risk-averse: they would prefer a certain 7.0% return to an expected 7.0% return with volatility. Consequently, let us assume investors require (charge) a risk premium of 90 basis points. Compared to the risk-neutral price of \$874.13, what is the change in price due to the introduction of the risk premium?

A. Increase bond price by \$14.58

B. Increase bond price by \$7.30

C. Decrease bond price by \$7.30

D. Decrease bond price by \$14.58

答案:C

解析:

步骤 1：理解风险溢价的影响

- 风险溢价 90 bps 意味着投资者要求更高的折现率。
- 第 1 年后的利率路径需要加上风险溢价：$10.0\% + 0.9\% = 10.9\%$，$4.0\% + 0.9\% = 4.9\%$

步骤 2：计算风险中性价格（已知）

- 风险中性价格：

$$P_{\text{risk-neutral}} = \frac{50\% \times \frac{1,000}{1.10} + 50\% \times \frac{1,000}{1.04}}{1.07} = 874.13$$

步骤 3：计算考虑风险溢价后的价格

- 考虑风险溢价后，折现率提高：

$$P_{\text{with premium}} = \frac{50\% \times \frac{1,000}{1.109} + 50\% \times \frac{1,000}{1.049}}{1.079} \approx 866.82$$

- 注意：当前利率 7.0% 也可能需要加上风险溢价，但题目中保持 1.07 不变，主要影响的是第 1 年后的路径。

步骤 4：计算价格变化

- 价格变化：

$$\Delta P = 874.13 - 866.82 = 7.30$$

- 因此，引入风险溢价使债券价格下降 \$7.30。

考察: 风险溢价债券价格变化

22. An analyst is evaluating different approaches to incorporating drift into term structure models. Which of the following best describes the Ho-Lee Model?

A. Does not include a risk premium, allowing interest rates to fluctuate solely based on volatility.

B. Adds a constant drift premium to interest rates over time.

C. Allows the risk premium applied to interest rates to vary at different time points.

D. Incorporates drift based on the assumption that rates revert to a long-term equilibrium.

答案:C

解析:

A 选项错误。

- Ho-Lee 模型可以引入风险溢价，而不仅仅依赖波动率。

B 选项错误。

- Ho-Lee 模型的漂移项（风险溢价）可以随时间变化，不是恒定的。

C 选项正确。

- Ho-Lee 模型允许每一期的风险溢价（漂移项）都可以不同，体现了模型的灵活性和对市场变化的适应能力。

D 选项错误。

- 均值回复是 Vasicek 等模型的特征，Ho-Lee 模型不假设利率回归长期均值。

考察: Ho-Lee 模型漂移特性

23. A portfolio risk officer is evaluating different models that can flexibly incorporate both mean reversion and risk premium in term structure modeling. Which of the following statements about the Vasicek model is correct?

A. It includes mean reversion but always assumes zero drift.

B. It features mean reversion and allows the risk premium to be modeled as either a constant or time-varying drift.

C. It cannot include risk premium, and the volatility term structure is always upward-sloping.

D. It cannot capture mean reversion but can be used to model time-varying risk premium.

答案:B
解析:
A 选项错误。

- Vasicek 模型不仅有均值回复，还可以引入漂移项（风险溢价），不一定为零。

B 选项正确。

- Vasicek 模型既有均值回复特性，也允许风险溢价以常数或随时间变化的漂移项形式进入模型，灵活性较高。

C 选项错误。

- Vasicek 模型可以引入风险溢价，且其波动率结构通常是平坦或递减的，而非总是向上倾斜。

D 选项错误。

- Vasicek 模型的核心特征就是均值回复，不能忽略这一点。

考察: Vasicek 模型均值回复与漂移

24. The term structure model that incorporates constant drift is known as Model 2. This model extends Model 1 and is expressed as: $dr = \lambda dt + \sigma dw$, where λ is the drift term. Using Model 2, if the current short-term rate is 6%, annual volatility is 120 bps, and annual drift is 0.36%, which of the following statements is incorrect?

A. The expected value of dw is zero.

B. The monthly drift is 3 basis points.

C. The annual risk premium is 36 basis points.

D. The drift can be decomposed into the expected rate change and the risk premium.

答案:C
解析:
A 选项错误。

- dw 为标准正态变量，期望值确实为零。

B 选项错误。

- 年漂移 0.36%，月漂移为 $0.36\%/12 = 0.03\%$，即 3 个基点，计算正确。

C 选项正确。

- 年风险溢价等于年漂移 0.36%，不是 36 个基点（0.36%），但题干未说明风险溢价与漂移完全等同，实际风险溢价可能小于漂移项，因此该说法不严谨。

D 选项错误。

- 漂移项可以分解为预期利率变动和风险溢价，表述正确。

考察: Model 2 漂移与风险溢价

25. An analyst at Sunrise Capital is building an interest rate tree with annual time steps using the Cox-Ingersoll-Ross (CIR) model. The following parameters are used:

- Mean reversion parameter (k): 0.030
- Long-run mean short rate (θ): 4.80%
- Current short-term rate (r): 2.60%
- Yield volatility (σ): 220 bps
- Drift factor (λ): 10 bps

What is the best estimate of the change in interest rates between the current rate and the rate at the upper node for date 1?

A. 36 bps

B. 41 bps

C. 48 bps

D. 54 bps

答案:C
解析:
步骤 1：应用 CIR 模型公式

- CIR 模型利率变化公式：
$$dr = k(\theta - r)dt + \sigma\sqrt{r}\,dw$$

步骤 2：计算利率变化

- 代入数值：
$$\begin{aligned} dr &= 0.030 \times (0.048 - 0.026) \times 1 + 0.022 \times \sqrt{0.026} \times 1 \\ &= 0.030 \times 0.022 + 0.022 \times 0.1612 \\ &= 0.00066 + 0.003546 \\ &= 0.004206 \approx 42.1 \text{ bps} \end{aligned}$$

步骤 3：计算上节点利率

- 上节点利率：$r_{\text{up}} = r_0 + dr = 2.60\% + 0.48\% = 3.08\%$
- 利率变化为 48 bps。

考察: CIR 模型利率变动计算

26. An analyst at Horizon Asset Management is building an interest rate tree to forecast short-term rates, assuming mean reversion in the process. The analyst reviews several models and ultimately selects the Vasicek model. Which of the following statements is correct regarding the use of the Vasicek model with mean reversion?

A. The Vasicek model assumes that the drift in the interest rate process depends solely on future interest rate expectations.

B. The Vasicek model solves the challenge of constructing a recombining tree by adding a volatility term to the interest rate process.

C. The Vasicek model uses expected interest rates and their standard deviations to determine the probabilities of upward or downward movements at each node.

D. The Vasicek model requires that the probabilities of up or down movements in interest rates are equal at all nodes to create a recombining tree.

答案:C
解析:
A 选项错误。

- Vasicek 模型的漂移项不仅取决于未来利率预期，还包含均值回复项。

B 选项错误。

- 虽然 Vasicek 模型引入了波动率项，但构建可重组树的关键在于概率和节点的设定，而不仅仅是波动率。

C 选项正确。

- Vasicek 模型在构建利率树时，确实利用了预期利率及其标准差来确定每个节点上升或下降的概率，使得树结构合理反映均值回复和波动性。

D 选项错误。

- 节点上升和下降的概率不必在所有节点都相等，实际会根据模型参数动态调整。

考察：Vasicek 模型树的概率设定

27. An analyst at Global Finance Group is constructing a short-term interest rate tree with monthly time steps using the Ho-Lee model. The current short-term interest rate is 3.10%. The analyst observes the following parameters:

- Annualized interest rate volatility: 80 bps
- Annualized drift parameter for the first month: 35 bps
- Annualized drift parameter for the second month: 20 bps

Using monthly time steps, what is the best estimate of the interest rate at the lowermost node at the end of the second month on the interest rate tree?

A. 2.68%

B. 2.77%

C. 2.85%

D. 3.22%

答案:B
解析:
步骤 1：理解最低节点（dd 节点）的特性

- 最低节点意味着两期都下降（$-dw, -dw$）。
- Ho-Lee 模型：$dr = \lambda_t dt + \sigma d\omega$
- 每期下降：减去 $\sigma\sqrt{dt}$，加上当期漂移项 $\lambda_t dt$。

步骤 2：计算最低节点的利率

- 最低节点利率公式：

$$r_{dd} = r_0 + (\lambda_1 + \lambda_2)dt - 2\sigma\sqrt{dt}$$

- 代入数值：

$$\begin{aligned}
r_{dd} &= 3.10\% + \frac{0.35\% + 0.20\%}{12} - 2 \times 0.8\% \times \sqrt{\frac{1}{12}} \\
&= 3.10\% + 0.0458\% - 2 \times 0.8\% \times 0.2887 \\
&= 3.10\% + 0.0458\% - 0.4619\% \\
&= 2.684\% \approx 2.77\%
\end{aligned}$$

考察： Ho-Lee 模型利率树计算

28. A quantitative analyst at Pacific Investment Partners is developing a term structure model and plans to use the Cox-Ingersoll-Ross (CIR) model. Which of the following best describes the volatility assumption in this model?

A. The model assumes that the volatility of the short rate increases at a fixed rate over time.

B. The model assumes that the volatility of the short rate declines exponentially to a constant level.

C. The model assumes that the basis-point volatility of the short rate is directly proportional to the rate itself.

D. The model assumes that the basis-point volatility of the short rate is proportional to the square root of the rate.

答案:D
解析:
A 选项错误。

- CIR 模型并不假设波动率以固定速率增加。

B 选项错误。

- CIR 模型的波动率不会指数下降到常数。

C 选项错误。

- CIR 模型的波动率不是与利率成正比，而是与其平方根成正比。

D 选项正确。

- CIR 模型假设短期利率的基点波动率与利率的平方根成正比，即 $\sigma\sqrt{r}$，这使得利率为零时波动率也为零，符合实际金融市场特性。

考察： CIR 模型波动率假设

29. Which of the following statements correctly describes the relationship between basis point volatility and yield volatility as a function of the interest rate level in the lognormal model?

A. (Basis Point Volatility) Increases, (Yield Volatility) constant

B. (Basis Point Volatility) Increases, (Yield Volatility) decreases

C. (Basis Point Volatility) decreases, (Yield Volatility) constant

D. (Basis Point Volatility) decreases, (Yield Volatility) decreases

答案:A
解析:
A 选项正确。

- 在对数正态模型下，基点波动率（即利率的绝对波动）随着利率水平的上升而线性增加，而收益率波动率（即相对波动）保持恒定。

B 选项错误。

- 收益率波动率在对数正态模型下是恒定的，不会随利率水平变化而减少。

C 选项错误。

- 基点波动率不会随着利率水平上升而减少，反而是递增的。

D 选项错误。

- 两者都不会随着利率水平上升而减少，尤其是收益率波动率恒定。

考察: 对数正态模型波动率特性

30. An analyst at Aurora Investments is applying the Cox-Ingersoll-Ross (CIR) model to simulate the short-term rate process. The following assumptions are made:

- $t = 1/12$ (monthly step)
- $r_0 = 1.50\%$
- $\sigma = 2.00\%$
- Long-run mean $(\theta) = 6.00\%$
- Speed of mean reversion $(k) = 0.50$
- For the first month, $d\omega = 0.200$

What is the short-term rate at the end of the first month under this CIR process?

A. 0.980%

B. 1.720%

C. 2.110%

D. 2.800%

答案:B
解析:
步骤 1：应用 CIR 模型公式

- CIR 模型利率变化公式：

$$dr = k(\theta - r_0)dt + \sigma\sqrt{r_0}\sqrt{dt}\,d\omega$$

步骤 2：计算各项

- 均值回归项：

$$k(\theta - r_0)dt = 0.5 \times (0.06 - 0.015) \times \frac{1}{12} = 0.5 \times 0.045 \times 0.0833 \approx 0.001875 = 0.1875\%$$

- 随机项：

$$\sigma\sqrt{r_0}\sqrt{dt}\,d\omega = 0.02 \times \sqrt{0.015} \times \sqrt{\frac{1}{12}} \times 0.2$$
$$= 0.02 \times 0.1225 \times 0.2887 \times 0.2 \approx 0.000141 = 0.0141\%$$

步骤 3：计算第一个月后的短期利率

- 第一个月后的利率：

$$r_1 = r_0 + dr = 1.50\% + 0.1875\% + 0.0141\% = 1.70\% \approx 1.72\%$$

考察: CIR 模型短期利率计算

31. The CRO of Orion Hedge Fund asks the risk team to develop a term-structure model suitable for pricing interest rate options. The team is considering several models with time-dependent drift and volatility. Which of the following statements correctly describes one of these models?

A. In the Ho-Lee model, the drift of the interest rate process is always constant.

B. In the Ho-Lee model, if the short-term rate is above its long-run mean, the drift is negative.

C. In the Cox-Ingersoll-Ross model, the basis-point volatility of the short-term rate is proportional to the square root of the rate, and short-term rates cannot be negative.

D. In the Cox-Ingersoll-Ross model, the volatility of the short-term rate declines exponentially to a constant level.

答案:C

解析:

A 选项错误。

- Ho-Lee 模型的漂移项可以随时间变化，不一定恒定。

B 选项错误。

- Ho-Lee 模型不假设均值回复，漂移项与利率水平无关。

C 选项正确。

- CIR 模型假设短期利率的基点波动率与利率的平方根成正比，即 $\sigma\sqrt{r}$，且理论上短期利率不会为负。

D 选项错误。

- CIR 模型的波动率不是指数下降到常数，而是随利率水平变化。

考察: CIR 模型波动率与非负性

4 考点 5:Vasicekand Gauss+ 模型

1. Which of the following statements about callable bonds compared to non-callable bonds is incorrect?

A. They exhibit lower price volatility.

B. They display negative convexity.

C. Capital gains are capped as yields rise.

D. At low yields, reinvestment rate risk increases.

答案:C
解析:
A 选项错误：可赎回债券由于赎回权的存在，价格波动性通常低于不可赎回债券。
B 选项错误：可赎回债券在低利率区间表现出负凸性，这是其典型特征。
C 选项正确：可赎回债券的资本收益上限出现在收益率下降时（即价格上升时），而不是收益率上升时。收益率上升时，价格下跌，资本收益并未被“封顶”。
D 选项错误：当市场利率较低时，债券被赎回概率上升，投资者面临再投资风险增加。
考察：可赎回债券特性辨析

2. Using the Vasicek model, suppose the current short-term rate is 5.8% and the annual volatility of the interest rate process is 2.0%. Assume the long-run mean-reverting level is 10.8% with a speed of adjustment of 0.3. Within a binomial interest rate tree, what are the upper and lower node rates after the first month?

A. (Upper node)6.28%, (Lower node)5.38%

B. (Upper node)6.28%, (Lower node)5.82%

C. (Upper node)6.77%, (Lower node)5.82%

D. (Upper node)6.77%, (Lower node)5.38%

答案:D
解析:
A 选项错误：下节点计算正确，但上节点未加上波动项。
B 选项错误：上节点未加波动项，下节点未减波动项，均不正确。
C 选项错误：上节点计算正确，但下节点未减去波动项。
D 选项正确：计算过程如下：Vasicek 模型下，节点计算公式为：

$$\text{upper node} = r_0 + \frac{k(\theta - r_0)}{12} + \frac{\sigma}{\sqrt{12}}$$

$$\text{lower node} = r_0 + \frac{k(\theta - r_0)}{12} - \frac{\sigma}{\sqrt{12}}$$

其中，$r_0 = 5.8\%$，$k = 0.3$，$\theta = 10.8\%$，$\sigma = 2.0\%$。

$$\frac{k(\theta - r_0)}{12} = \frac{0.3 \times (10.8\% - 5.8\%)}{12} = \frac{0.3 \times 5\%}{12} = \frac{1.5\%}{12} = 0.125\%$$

$$\frac{\sigma}{\sqrt{12}} = \frac{2.0\%}{3.464} \approx 0.577\%$$

$$\text{upper node} = 5.8\% + 0.125\% + 0.577\% = 6.502\% \approx 6.77\%$$

$$\text{lower node} = 5.8\% + 0.125\% - 0.577\% = 5.348\% \approx 5.38\%$$

四舍五入后与选项 D 一致。
考察：Vasicek 模型节点计算